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MEng Project Report
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**Quantum Design and
Simulation of MASERs in
the Strong Coupling Limit**

Abstract

This project investigates whether the shape of a dielectric resonator could be optimised to maximise the Purcell factor of a solid-state MASER at room temperature and zero field. COMSOL Multiphysics® and a Java-based genetic algorithm (GA) were used to simulate and test thousands of resonator shapes to determine those that optimised electromagnetic confinement while still being resonant at 1.45 GHz. A 10x10 binary subgrid design using a binary matrix allowed for shape exploration without remeshing. Results showed that in theoretical, loss-free conditions, the optimised geometry significantly outperformed the original MASER-in-a-Shoebox design in Purcell factor and simulated masing output power. However, when the dielectric is assigned a realistic loss tangent, this outperformance was not seen and the GA could not find dielectric shapes which operate at the correct mode. Additional future work would require exploring transfer to COMSOL LiveLink™ for MATLAB®, which would result in more robust algorithms and co-localisation with machine learning methods. Overall, this work reassures that resonator geometry has a strong influence on MASER performance and provides a foundation for more advanced, physically realistic optimisation of cavity geometries for future microwave technologies.

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AI declaration

Artificial Intelligence (AI) tools were used to support the research process by clarifying concepts that were initially difficult to understand and assisting in diagnosing errors in code, which were subsequently corrected by myself. During the writing process, AI was also employed to help identify and correct grammatical and punctuation errors. All other work is my own unless otherwise stated. ChatGPT-3.5, OpenAI, url: 'https://openai.com/'

Collaboration and supervision

When starting this project, I had no prior experience in the field of microwave engineering, nor had I undertaken a research project that involved coding. The only knowledge I arrived with was from courses (Optoelectronics and Maths & Computing) I had taken throughout my degree.

Acknowledgements

I would like to thank Prof. Mark Oxborrow, my supervisor, for his guidance through this project. His ability to explain complex topics with clarity and ease made a significant difference in the initial understanding of this research field. I would also like to express my gratitude to everyone who waited patiently for my simulations to finish before using the COMSOL® Floating Network Licence, your patients was very appreciated.

Chapter 1

Aims and Context

A solid-state MASER operable at room temperature and zero magnetic field has recently been demonstrated in a compact, portable form factor. While this represents a significant step toward practical implementation, low energy efficiency, specifically the limited output power relative to the high optical pump input, continues to constrain real-world adoption of the technology.

This project focuses on the design and optimisation of dielectric resonators used within the MASER cavity, with the aim of enhancing the Purcell factor and thereby improving masing efficiency. A genetic algorithm (GA) is employed in conjunction with COMSOL Multiphysics to evolve resonator geometries that maximise the Purcell factor by optimising quality factor (Q_f) and mode volume (V_m). The hypothesis is that irregular resonator shapes can offer superior confinement and lower radiative loss than traditional cylindrical designs, even when subject to practical material constraints.

This research is situated within the broader context of low-noise microwave amplification, a technology with wide-reaching applications in quantum information systems, deep-space communication, and highly sensitive spectroscopy. Improving MASER efficiency at room temperature has the potential to remove key barriers to deployment in industry. As quantum and communication technologies continue to develop, efficient room-temperature MASERs could form a foundational component in next-generation sensing and amplification platforms.

Chapter 2

Statement on effectiveness of your own performance and the team performance

A large portion of this project required independent work, particularly in the integration of COMSOL Multiphysics with a genetic algorithm written in Java. I was solely responsible for designing and executing the simulation workflow, which involved building a programmatic interface using the COMSOL Java API, developing and tuning the genetic algorithm, and handling large volumes of simulation data. This independence meant I had to manage my time efficiently, troubleshoot errors without immediate supervision, and learn new technical skills on my own, including how to script mesh-free geometry updates, automate parametric sweeps, and extract electromagnetic field data for postprocessing. This project gave me the opportunity to demonstrate and refine my problem-solving skills. I had to carefully isolate variables, run controlled tests, and develop practical workarounds. This trial-and-error approach deepened my understanding of both the simulation tools and the underlying physics. Although the technical work was done mostly independently, collaboration was still important. Early discussions with post-graduates helped me get set up with the software and avoid common pitfalls. I benefited especially from informal peer discussions which helped me learn valuable debugging strategies, simulation efficiency tricks. Weekly supervisor meetings in the first term were especially useful for developing the foundational understandings of MASERs and helped keep the project on track.

Chapter 3

Literature Review

3.1 Introduction to MASERs

MASER is an acronym for Microwave Amplification by Stimulated Emission of Radiation. It is closely related to the LASER (Microwave being simply replaced with Light), although MASERs actually predate LASERs[16]. While LASERs produce coherent beams of visible or infrared light, MASERs amplify coherent microwave signals. They share the same underlying physical principle, but MASERs operate at much lower frequencies and differ in design[1], [2]. A typical MASER consists of a gain medium (often referred to as a masing crystal), optical pump, metal cavity, and dielectric resonator[3]–[5]. This set up can be seen in Figure 3.1.1 which shows the basic anatomy of a MASER.

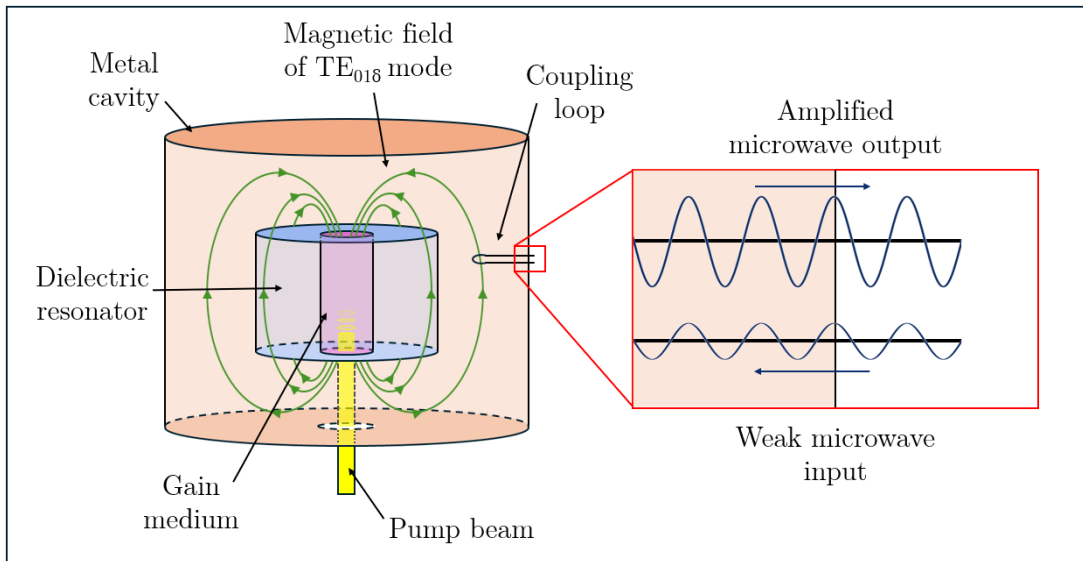


Figure 3.1.1: The typical anatomy of a room temperature room, with all major features labelled. An RF field is induced, the amplified microwave signal is output via the coupling.

The generation of a coherent microwave output (masing) relies on the transition of atoms and molecules in the gain medium between its discrete energy levels. When a microwave interacts with this medium, it results in the emission of additional photons with the same frequency, phase and direction. This phenomenon is called stimulated emission[1], [2]. Stimulated emission occurs when an incoming photon interacts with an atom or molecule in an excited state resulting in the emission of a second photon with identical properties. For continual stimulated emission, there must be more atoms or molecules in excited states than in lower states. This is called a population inversion, shown in Figure 3.1.2(b), and it ensures that incoming microwaves achieve stimulated emission rather than being absorbed by the masing crystal. A population inversion is created by supplying energy to the gain medium (pumping), hence the need for an optical pump[6]. Controlled pumping is essential for MASER operation[3], [4]. A simple Jablonski diagram in Figure 3.1.2(a) shows the excitation of atoms using a pump beam.

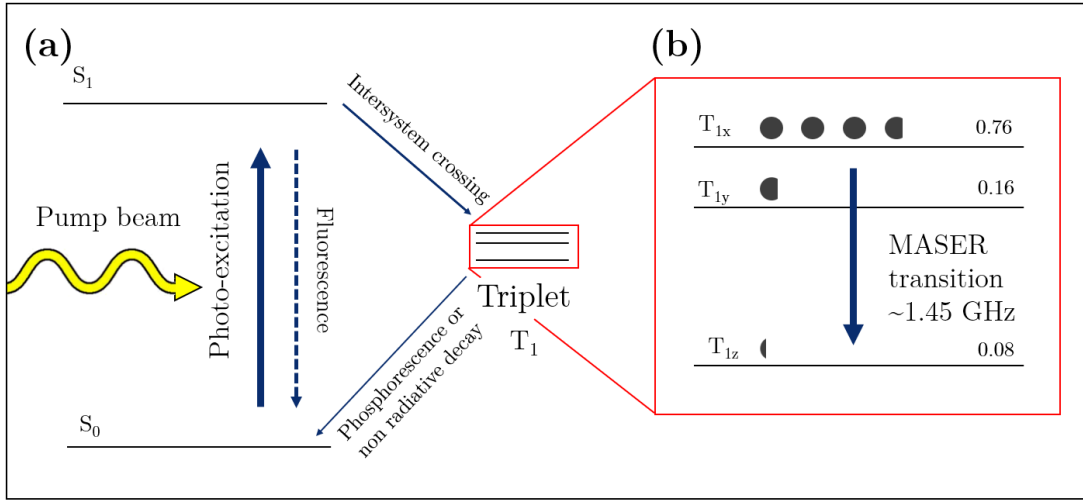


Figure 3.1.2: A Jablonski diagram, with (a) representing photo-excitation from the ground to first excited state by a pump beam and (b) the establishment of population inversion as T_{1x} possesses 76% of atoms and T_{1z} possesses only 8%.

Role of the Microwave Cavity and Gain Medium

Microwaves inside the metal cavity form a microwave electromagnetic field (with the magnetic field visible in Figure 3.1.1). The field consists of oscillating electric and magnetic components at a frequency typically in the gigahertz (GHz) range[7]. These fields resonate in specific spatial patterns, or cavity modes, determined by the cavity geometry and boundary conditions. The $TE_{01\delta}$ mode is most commonly used in MASER designs due to its high magnetic field concentration in the dielectric region, although other modes such as the anapole mode have also been proposed for their potential to reduce radiative losses and enhance confinement. The electric field confinement of these modes is shown in Figure 3.1.3. The gain medium interacts with this electromagnetic field; stimulated emission is driven by this interaction[3], [4]. The degree to which energy is transferred from the gain medium to photons depends on how strongly the gain medium couples to the cavity's field[4], [8].

Electric field normal [$V\ m^{-1}$]

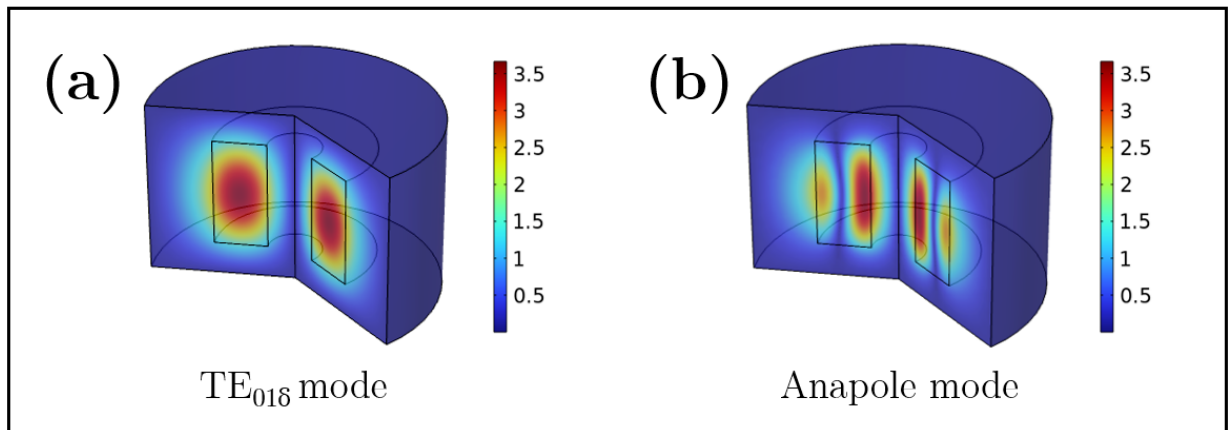


Figure 3.1.3: Comparison of simulated electric field normal components in COMSOL for (a) the $TE_{01\delta}$ mode and (b) the anapole mode. Both modes are shown within identical cylindrical cavities to illustrate differences in field distribution.

The system is said to enter the strong coupling regime when the rate of energy exchange between the gain medium and the cavity photons exceeds the rate at which energy is lost from the system. Crossing this threshold, known as the strong coupling limit, enables the MASER to operate with higher efficiency and greater stability[2], [4], [9]. The strength of this interaction is quantified by the spin-photon coupling strength g , which represents the rate of coherent energy exchange between the cavity field and the spins in the gain medium[10]. This coupling strength is enhanced by increasing both the spatial overlap between the

magnetic field and the gain medium, and the density of active spins[2], [4]. In most MASER implementations, a coupling loop or antenna is used to extract microwave power from the cavity or inject signals into it[3], [11].

$$g = \frac{\mu \cdot E(\mathbf{r}_0)}{\hbar} = \frac{\mu}{\hbar} \sqrt{\frac{\hbar \omega}{2\varepsilon_0 \varepsilon(\mathbf{r}_0) V_m}} \quad (3.1.1)$$

In this expression, g is the coupling rate (Hz), μ is the transition dipole moment of the emitter, $E(\mathbf{r}_0)$ is the electric field at the emitter location $\mathbf{r}_0$, ω is the angular frequency of the cavity mode, ε_0 is the vacuum permittivity, $\varepsilon(\mathbf{r}_0)$ is the relative permittivity at the emitter's position, V_m is the mode volume, and $\hbar$ is the reduced Planck constant.

Dielectric Resonators

Another way to increase the number of photons produced, other than increasing the coupling, is to increase the energy transferred into the gain medium. This can be achieved by confining the oscillating magnetic field within the gain medium. This is the role of the dielectric resonator[3], [12]. The confinement of the field plays a key role in determining the efficiency of energy transfer within the device and is measured by V_m .

$$V_m = \frac{\int \mu(\mathbf{r}) |\mathbf{H}(\mathbf{r})|^2 d^3\mathbf{r}}{\mu(\mathbf{r}_0) |\mathbf{H}(\mathbf{r}_0)|^2} \quad (3.1.2)$$

In this expression, $\mu(\mathbf{r})$ is the permeability at position $\mathbf{r}$, $\mathbf{H}(\mathbf{r})$ is the magnetic field distribution of the mode, and $\mathbf{r}_0$ denotes the position of maximum magnetic energy density.

To increase the confinement of the field (and thereby reduce V_m) a material with a high relative permittivity, ε_r , should be used. Appendix Table A.1 shows a comparison of the most frequently used materials in dielectric resonator design.

The dielectric resonator is therefore a crucial component in a MASER, but also the cavity needs to be well designed to minimise system losses. Losses occur through many mechanisms with the main types being dielectric losses (intrinsic property of all materials, $\tan \delta$)[13], [14], radiative losses (energy escaping the cavity), and conductive losses (energy dissipated in cavity walls)[15], [16]. Masing occurs when the gain provided by the inverted spin population exceeds the total cavity losses, defining the masing threshold condition[17]. Figure 3.1.4 shows the difference in V_m between a cavity with and without a dielectric resonator.

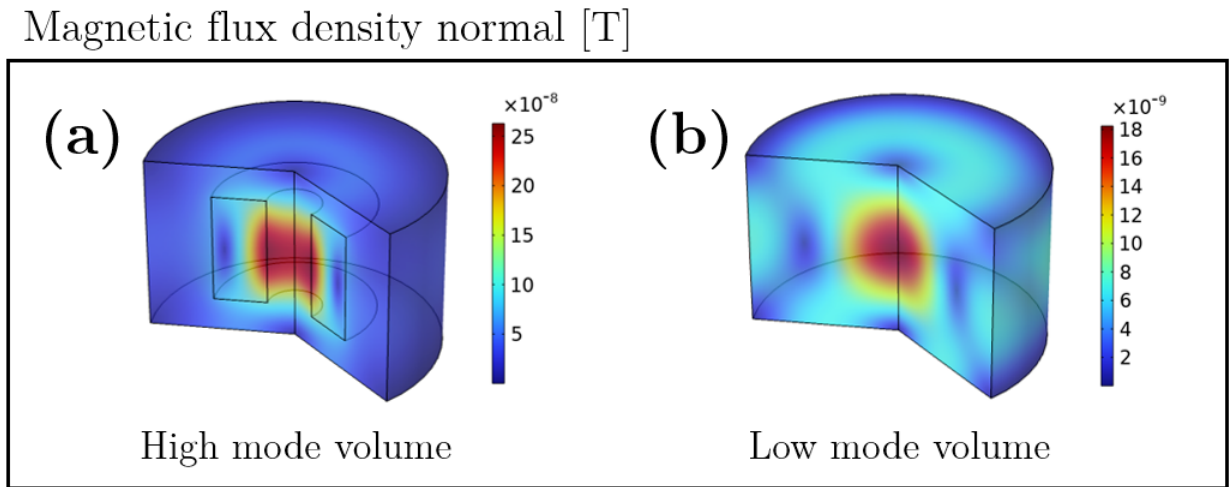


Figure 3.1.4: Magnetic flux density normal plots from COMSOL comparing (a) a cavity with a dielectric resonator, resulting in high mode volume, and (b) a cavity without a dielectric resonator, thus showing reduced mode volume due to a reduction in field confinement.

3.2 Recent Developments

The first successful operation of a room-temperature solid-state MASER in 2012 marked a significant breakthrough in the field. This device utilised a pentacene-doped p-terphenyl crystal as the gain medium, optically pumped by a pulsed dye laser emitting at 590 nm[3]. The system achieved stimulated emission at approximately 1.45 GHz without the need for cryogenic cooling or external magnetic fields. The pentacene molecules, upon optical excitation, underwent intersystem crossing to a triplet state, leading to a population inversion essential for masing[3], [18].

Despite this advancement, the initial design faced challenges. The MASER operated only in pulsed mode, with high optical pumping power requirements (~ 1.4 kW), and suffered from significant dielectric and radiative losses within the cavity, limiting its efficiency and practical applications[4], [18], [19].

Subsequent research focused on improving dielectric resonator designs to enhance MASER performance. High-permittivity materials such as strontium titanate (SrTiO_3) and titanium dioxide (TiO_2) were explored to reduce mode volume and increase the Purcell factor, thereby strengthening the coupling between the cavity field and the gain medium[20]–[22]. Innovations in resonator geometries, including the use of whispering-gallery modes and photonic crystal structures, have been investigated to achieve higher quality factors and better field confinement[23]–[26].

Efforts to identify alternative gain media led to the exploration of diazapentacene-doped p-terphenyl, which demonstrated improved chemical stability and lower masing thresholds compared to pentacene-based systems. Additionally, nitrogen-vacancy (NV) centers in diamond have been proposed as potential room-temperature gain media due to their long spin coherence times and compatibility with continuous-wave operation[5], [27]–[29].

Recent developments have aimed at miniaturizing MASER systems for practical applications. The "MASER-in-a-Shoebox" project successfully demonstrated a portable, plug-and-play MASER device operating at room temperature and zero magnetic field, highlighting the potential for MASER technology in real-world scenarios[30].

Other research has explored various dielectric resonator geometries to enhance MASER performance. Traditional cylindrical and rectangular shapes have been complemented by innovative designs such as hemispherical, hexagonal, conical, and trapezoidal forms[12]. These geometries aim to optimize field confinement and mode distribution, thereby improving the resonator's quality factor and reducing mode volume[23], [24]. For instance, asymmetric cross-shaped dielectric resonators have demonstrated wideband circular polarization, which can be beneficial in certain MASER applications.

3.3 The Purcell Factor

Dielectric resonators serve a critical function in MASERs by concentrating the magnetic flux within the gain medium, thereby enhancing the interaction between the electromagnetic field and the active material. This concentration is pivotal for achieving the strong coupling regime necessary for efficient MASER operation[3], [9]. The geometry and material properties of the resonator directly influence the distribution and intensity of the electromagnetic fields, affecting the device's overall performance[23]. The performance of a dielectric resonator in a MASER is fundamentally determined by two interrelated metrics: the quality factor (Q_f) and V_m . The quality factor measures how long energy is stored in the resonator before it is lost, whether through dielectric absorption or any other losses in the system[31]. A high Q_f value implies low intrinsic loss, enabling the resonator to sustain strong electromagnetic fields over longer periods. Figure 3.3.1 demonstrates the dissipation of energy with time in a high and low quality factor resonator. This sustained field is critical for facilitating multiple interactions between the gain medium and the cavity field, which increases the probability of stimulated emission[32]. The mode volume has already been defined and can be seen in Eq 3.1.2.

These two parameters combine to define the Purcell factor (F_p), a measure of how much the cavity enhances the spontaneous emission rate of the gain medium. The Purcell factor is given by:

$$\mathbf{F_P} = \frac{3}{4\pi^2} \left(\frac{\lambda_{\text{free}}}{n} \right)^3 \frac{Q}{V_m} \quad (3.3.1)$$

where λ_{free} is the free-space wavelength of the emitter, and n is the refractive index of the medium.

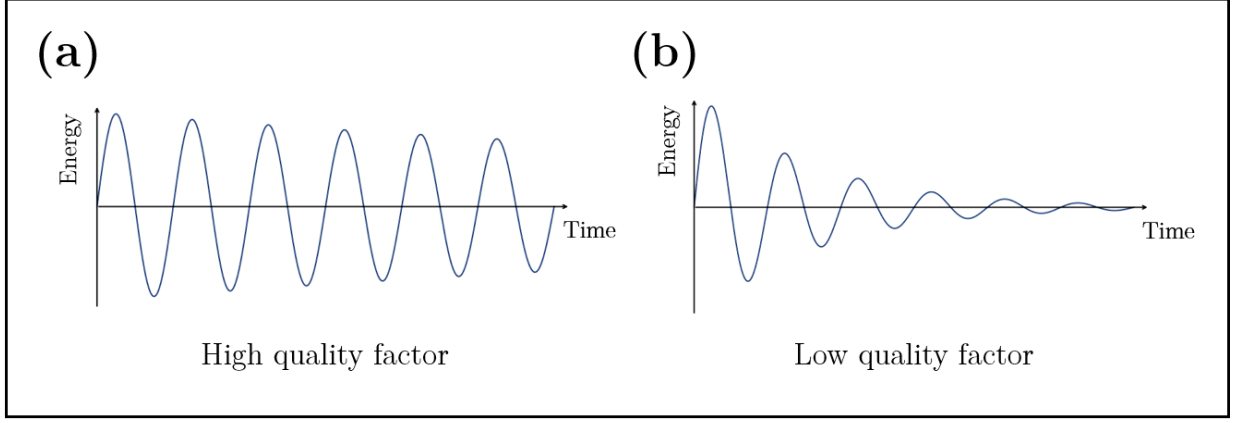


Figure 3.3.1: Graphical comparison of (a) high quality factor and (b) low quality factor resonators, showing how energy decays over time. High- Q systems retain energy longer with slower decay, while low- Q systems dissipate energy rapidly.

In the context of MASERs, a high Purcell factor implies enhanced emission and increased likelihood of reaching the masing threshold, especially in systems with limited pump power or low intrinsic gain.

Recent studies have demonstrated that by carefully tailoring dielectric resonator geometries, such as incorporating notches, split-rings, or using hybrid photonic structures, both Q_f and V_m can be optimised simultaneously[24]. For example, Rybin et al. showed that supercavity modes in subwavelength dielectric resonators can yield ultra-high Q factors by exploiting interference between leaky and non-leaky modes to trap energy more effectively[23]. Similarly, broken-symmetry designs have been used to excite Fano resonances, leading to sharp spectral features with high Q_f and relatively small V_m , offering a practical route to maximise the Purcell enhancement in compact geometries[24].

A key related parameter is the cooperativity, C , which quantifies the ratio of coherent coupling to dissipation in the system [10], [28]. It is often expressed as:

$$C = \frac{g^2}{\kappa\gamma} \quad (3.3.2)$$

where g is the spin-photon coupling strength, κ is the cavity decay rate, and γ is the spin dephasing rate.

Since a high Purcell factor enhances the interaction strength between the cavity field and the gain medium, it directly contributes to increasing cooperativity. High cooperativity is essential for efficient masing and is closely associated with low noise output, making it a critical for practical MASER applications in sensing and communication systems[28].

3.4 COMSOL Modelling of Resonators

In the design and optimisation of MASERs, a software is required that can accurately model the electromagnetic fields within dielectric resonators. COMSOL Multiphysics®, a finite element analysis software, is widely utilized for this purpose due to its capability to simulate complex geometries and material properties[33]–[35]. By solving Maxwell’s equations, COMSOL enables the visualisation and analysis of electric $\vec{E}$ and magnetic $\vec{H}$ field distributions within resonators. This is essential for determining the Q_f and V_m [36]. In COMSOL, Q can be calculated by integrating the electromagnetic energy density over the resonator volume and the power loss over its surface[48,50]. This approach has been demonstrated in COMSOL’s application examples, where models of rectangular, cylindrical, and spherical cavities are analysed to compute resonant frequencies and Q_f [35]. Additionally, COMSOL’s eigenfrequency analysis allows for the determination of resonant modes and their corresponding field distributions, allowing for the identification of modes with optimal Q_f/V_m ratios.

$$Q_f = \frac{1}{\tan \delta} = \frac{\epsilon'}{\epsilon''} \quad (3.4.1)$$

Where Q_f is the quality factor due to dielectric losses, $\tan \delta$ is the loss tangent of the material defined as $\tan \delta$, ϵ' is the real part of the complex permittivity, and ϵ'' is the imaginary part (loss component) of the complex permittivity.

V_m quantifies the spatial confinement of the resonant mode and is calculated by integrating the energy density over the entire volume and normalizing it to the maximum energy density within the resonator. COMSOL's capability to model complex resonator geometries, including whispering-gallery modes and photonic crystal structures, enables the exploration of designs that achieve high Q-factors and low mode volumes.

Additionally, COMSOL has been employed in the simulation of circuit quantum electrodynamics (cQED) systems, where superconducting qubits interact with microwave resonators. In these applications, precise modelling of $\vec{E}$ and $\vec{H}$ fields is essential for understanding light-matter interactions at the quantum level and for designing resonators with desired coupling strengths and coherence times[25].

3.5 Gaps in Current Literature and Motivation for This Project

Despite extensive progress in MASER development, there remains a notable gap in the literature regarding the systematic optimisation of dielectric resonator shapes and sizes. While several studies have investigated material properties and mode confinement within standard geometries such as cylindrical or spherical resonators[12], none have explored beyond these conventional designs to determine how variations in shape could influence the quality factor, mode volume, and, by extension, the Purcell factor. Although simulation tools such as COMSOL Multiphysics® have been employed to model specific dielectric resonators in MASER and photonic systems, the parameter space of resonator geometry itself remains largely unexplored in terms of global optimisation[37], [38].

To address this gap, the present project focuses on optimising the size and shape of dielectric resonators using the COMSOL API in conjunction with a genetic algorithm (GA). The goal is to maximise the Purcell factor by enhancing electromagnetic confinement and improving Q_f/V_m [13,21]. This approach allows for automated exploration of complex geometric configurations that may not be intuitive or easily accessible through manual tuning[37]–[39]. By treating the resonator geometry as a set of evolutionary parameters within the GA, the simulation framework identifies structures that yield superior field confinement and reduced radiative and dielectric losses[40], [41]. Beyond the scope of this project, future work could investigate integrating machine learning models to predict resonator performance or applying this methodology to other active materials and frequency bands could further extend the impact of this optimisation framework within both MASER and broader photonic technologies. While this study focuses on simulation-based optimisation, future work could also include experimental fabrication and testing of high-Purcell resonator designs to validate model predictions[9], [30].

3.6 Summary

MASERs (Microwave Amplification by Stimulated Emission of Radiation) operate on the same fundamental principle as lasers but emit coherent radiation in the microwave regime. Initially constrained by the need for cryogenic environments and external magnetic fields, a major milestone was achieved in 2012 with the demonstration of a room-temperature MASER using pentacene-doped p-terphenyl as the gain medium within a dielectric-loaded cavity. This system emitted at 1.45 GHz, but despite its breakthrough status, it remained limited to pulsed operation and suffered from high optical pump power demands and significant losses due to cavity radiation and dielectric absorption.

Recent developments have focused on improving MASER efficiency and stability by enhancing the electromagnetic confinement offered by dielectric resonators. Innovations in resonator geometries, including non-standard shapes and photonic crystal structures, aim to improve the cavity's quality factor (Q_f) while reducing mode volume (V_m), thereby increasing the Purcell factor (proportional to Q_f/V_m), a critical measure of the system's emission enhancement. High-permittivity, low-loss materials such as SrTiO₃ and TiO₂ are commonly investigated to further improve performance. However, most studies to date have relied on conventional resonator geometries, with little attention paid to shape optimisation.

This project addresses that gap by using COMSOL Multiphysics® in combination with a genetic algorithm to automatically optimise the size and shape of dielectric resonators for maximum Purcell factor.

By simulating the magnetic field distributions and calculating Q_f and V_m , the algorithm searches for configurations that offer superior field confinement and reduced energy loss. The approach aims to identify unconventional resonator shapes that may provide substantial performance gains. Future work could extend this methodology through machine learning models, offering new directions in both MASER and photonic device design.

Chapter 4

Evaluation of Algorithmic Methods

The optimisation of dielectric resonator shape and size within MASER systems is a novel area of research, with no established methodologies in literature. Traditional MASER resonator designs tend to rely on standardised geometries such as toroids using squares or cylinders. This project aimed to explore a much broader design space, where the resonator geometry could be modified in complex and potentially non-intuitive ways to maximise performance. Given this objective, it was necessary to first evaluate a wide range of potential modelling strategies.

Due to the inherent axisymmetric structure of MASER cavities and to reduce computational cost, all resonator geometries in this project were modelled using COMSOL's 2D axisymmetric mode. This simplification assumes cylindrical symmetry around a central axis, which not only halves the dimensional complexity of the problem but also significantly reduces memory usage and simulation time. Figure 4.0.1 shows a 3D cross-section of a typical MASER, with a zoomed-in view of the 2D axisymmetric section used for simulation in COMSOL. This 2D slice represents the effective modelling domain, capturing the radial and vertical structure of the cavity. This unfortunately means that geometries with asymmetry or complex 3D features are unable to be modelled in this project. However, 3D modelling is possible in COMSOL (and many other electromagnetic simulation tools) so there is the possibility of exploring this in future.

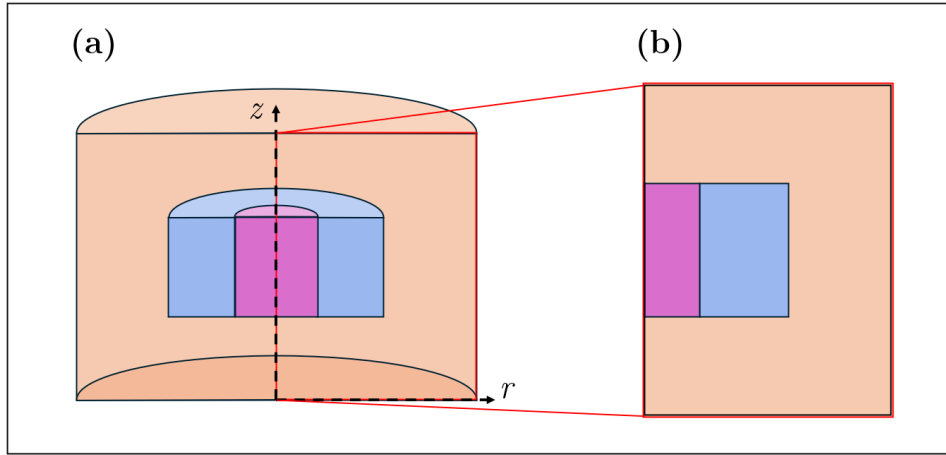


Figure 4.0.1: (a) 3D schematic of a typical MASER cavity showing the cylindrical symmetry of the structure. (b) Zoomed-in cross-section illustrating the 2D axisymmetric view used for COMSOL modelling, where only the r - z plane is simulated.

All these models face the challenge of retaining the resonant frequency of 1.45 GHz. Scaling the global geometries either up or down depending on whether the frequency is too high or low is the simplest way to fix this problem. However, upon implementation into an optimisation algorithm that needs to run concurrent simulation, the model will need to be ‘remeshed’ every time, greatly increasing computational cost. To avoid this the global permittivity values (for both the dielectric and air) can be scaled by a factor, α . This is equivalent to scaling the geometry but without the need to remesh.

This section reviews the main modelling strategies that were considered at the outset of the project.

4.1 The Slice Method

In the first and simplest method, the inner wall of the dielectric resonator is constrained to follow the vertical outer boundary of the gain medium. This vertical line, defined over a prescribed axial interval, serves as a static inner contour. From a set of evenly spaced points along this vertical line, horizontal radial lines are projected outward. Each of these horizontal segments has a tunable length r_i , which determines the radial extent of the resonator at a given height. Connecting the endpoints of these vectors produces a continuous 2D shape representing the cross-section of the dielectric resonator. This is visually represented in Figure 4.4.1(a).

Despite being simple and having good granularity control (by simply increasing the number of horizontal segments), this method was ultimately not selected for final optimisation due to its inability to represent irregular shapes, such as dielectrics with overhangs or undercuts.

4.2 The Polygon Method

In this approach, the shape of the dielectric resonator is defined by a set of discrete vertices, each located within the spatial boundaries set by the cavity walls and the fixed gain medium. Each vertex is described by its cylindrical coordinate pair (r_i, z_i) , allowing complete geometric freedom within the permitted design region. These points are then connected sequentially; each point joined to its neighbours to form a closed, continuous 2D shape. This method is shown in Figure 4.4.1(b).

This approach offers maximal flexibility, as it does not assume any underlying symmetry or regular structure in the resonator. However, this method introduces several computational challenges. Firstly, the lack of inherent symmetry results in greater mesh complexity and longer simulation times in COM-SOL. Secondly, maintaining a valid, non-self-intersecting polygonal structure during optimisation requires additional geometric constraints. Therefore, this method was rejected for final optimisation implementation.

4.3 The Amoeba Method

The geometry is defined by a central point, aligned with the axis of symmetry in a cylindrical coordinate system. From this centre, a dense set of vectors is projected outward, each characterised by a length r_i and an angle θ_i from the horizontal axis. A sufficiently large number of these vectors (typically 50 or more) are used such that the connected endpoints form a closed curve with a high degree of smoothness. The result is shown in Figure 4.4.1(c) and resembles the contours of an amoeba, hence the name.

This approach offers a highly flexible parameter space, enabling the representation of complex, asymmetric geometries that may not be accessible through more rigid section- or grid-based methods. This method could also be combined with spline interpolation to further smooth the contour[67]. Despite its flexibility, it is unable to produce geometries with self-intersections or internal vacancies as each radial vector defines a single, continuous boundary. Features such as voids within the dielectric resonator or hooked extensions cannot be captured. Alongside this, the computational cost of this method is significantly greater than any other methods discussed. Hence, it was ultimately not selected.

4.4 The Binary Grid Method

The final modelling approach selected for this project was a binary subgrid method, wherein the dielectric resonator is represented as an $N \times N$ grid of square elements. With each square element corresponding to a discrete region that can either be filled with dielectric material or air, forming a binary matrix. The structure of this method can be seen in Figure 4.4.1(d) and is further explained in the Methods section.

This method was ultimately chosen for several reasons:

1. The binary matrix maps naturally onto genetic algorithms.
2. Non-intuitive (even disconnected) shapes can be realised.
3. The $N \times N$ subgrid can be scaled easily for coarser or finer resolutions.
4. The ratio of geometries remains constant (with permittivity scaling), so remeshing is unnecessary.

Although the pixelated nature of the grid limits resolution and enforces blocky transitions between dielectric and air, the method offers a balance between modelling flexibility and implementation simplicity, making it a practical and scalable choice for this project.

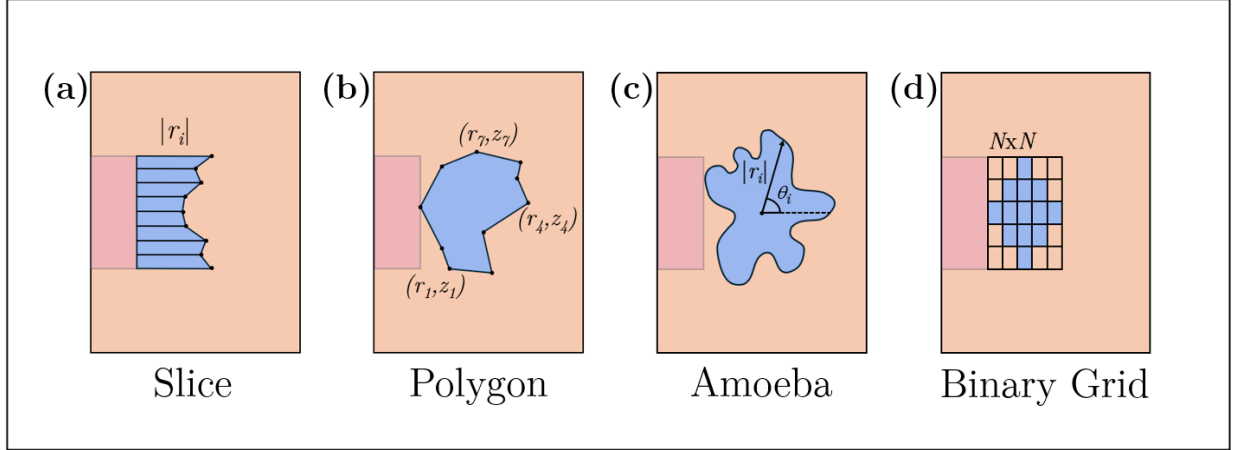


Figure 4.4.1: Visual comparison of four dielectric resonator modelling strategies. (a) The Slice method defines a shape using evenly spaced radial vectors of variable length r_i , projecting horizontally from a fixed vertical edge. (b) The Polygon method connects a series of coordinate points (r_i, z_i) to form a closed, angular 2D shape. (c) The Amoeba method creates a smooth, continuous outline using radial vectors defined by length r_i and angle θ_i , giving rise to curved geometries. (d) The Binary Grid method divides the dielectric region into an $N \times N$ matrix of square domains, each representing either air or dielectric, allowing for flexible pixel-based geometry control.

Chapter 5

Methods

5.1 COMSOL

All resonator geometries were modelled and solved using COMSOL Multiphysics® 6.0, a finite element analysis (FEA) software package designed for the simulation of complex physical systems. COMSOL uses edge element discretisation, which is well suited to solving Maxwell’s equations for electromagnetic wave propagation. This technique preserves the continuity of tangential electric field components across element boundaries, which is particularly important in high-frequency dielectric structures such as MASER resonators.

Due to the rotational symmetry of the resonator designs used in this project, all simulations were carried out using COMSOL’s ‘axisymmetric 2D mode’. This mode assumes azimuthal symmetry about the central axis and enables the reduction of a 3D problem to a 2D cross-section, significantly reducing computational time and memory requirements without sacrificing physical accuracy. In this simplified geometry, the dielectric resonator is modelled as a rectangular domain embedded in a rectangular air-filled cavity with metallic boundaries.

Each model comprised a dielectric resonator with a parametrised geometry and a surrounding air domain enclosed by a metal cavity. Copper was selected as the cavity material and modelled using an Impedance Boundary Condition to account for realistic surface losses. The surrounding air domain was included to prevent boundary reflections and to represent free-space propagation conditions. All materials were assigned appropriate electromagnetic properties, including relative permittivity, electric conductivity and relative permeability values. The gain medium was also approximated to have a relative permittivity equal to that of air, so it can simply be modelled as an air domain.

The initial dimensions of the resonator geometry were taken from the design reported in the “MASER-in-a-Shoebox” study by Ng et al.[30], Table 5.1. They demonstrated a compact, room-temperature, zero-field MASER. These baseline dimensions provided a realistic starting point for frequency scaling and optimisation. To facilitate flexible geometry control within COMSOL and ease integration with the optimisation process, the resonator model was constructed using piecewise regular domains. This approach allowed independent manipulation of features such as dielectric radius and the relative permittivity values, enabling efficient parameter updates.

To ensure model fidelity, the published dimensions from the MASER-in-a-Shoebox study were first replicated in COMSOL to serve as a geometric baseline. However, the original structure did not resonate precisely at the target frequency of 1.45 GHz. To correct this, small adjustments were made to the vertical dimensions z_1 and z_3 , effectively mimicking the practical tuning mechanism of a physical MASER resonator (like adjusting the cavity ceiling) to align the resonance with the desired frequency. After tuning, the model was confirmed to support the $TE_{01\delta}$ mode with strong magnetic field confinement in the dielectric region. This verified geometry was then used as a benchmark for evaluating all optimised configurations. Qualitative inspection of the field distributions also ensured that optimised geometries consistently exhibited the correct mode and avoided convergence onto non-masing solutions.

To enable control over the resonator’s shape during optimisation, a 10×10 rectangular subgrid was defined over the domain that contains the dielectric resonator. Each element of this subgrid represents a small region whose material properties can be independently adjusted. This complex structure would be time consuming to produce manually in COMSOL, so I wrote a code executable in COMSOL’s Application Builder to automate this process (can be found in Appendix B).

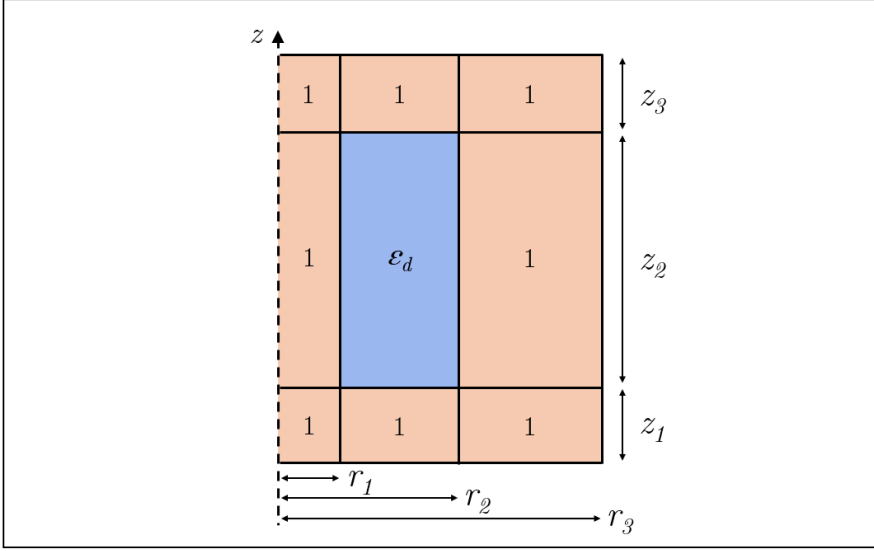


Figure 5.1.1: Schematic of the 2D axisymmetric model geometry used in simulations. The structure is divided into subdomains with defined radial (r_1, r_2, r_3) and vertical (z_1, z_2, z_3) dimensions. Each subdomain is assigned a relative permittivity, with air regions set to 1 and the dielectric core set to ϵ_d . This layout enables controlled manipulation of material boundaries and electromagnetic field confinement within the simulation domain.

Table 5.1: The input parameters of the baseline model, based off of the dimensions of the MASER-in-a-Shoebox[30]

Parameter	Value [mm]
z_1	2.60
z_2	8.70
z_3	2.60
r_1	2.05
r_2	6.10
r_3	11.0

By assigning either the relative permittivity of the dielectric material (see ϵ_r column in Table A.1) or that of air ($\epsilon_r=1$), each region can be treated as a binary switch, effectively functioning like a binary matrix embedded within the continuous geometry. This approach allows the overall shape of the resonator to be modified programmatically by toggling the permittivity in individual regions, enabling the genetic algorithm to explore a large variety of pixelated or composite geometries without redefining the full domain or rebuilding the mesh manually.

The primary goal of each simulation was to evaluate the magnetic field distribution $\vec{H}$ and the electric field distribution $\vec{E}$, from which the quality factor, Q_F , and mode volume, V_m , could be computed. Then therefore being able to calculate $\frac{Q_F}{V_m}$ which is proportional to F_P (the optimisation metric), as seen in Equation 3.3.1.

All models were meshed using extremely coarse triangular elements. This was to reduce simulation time as much as possible as many concurrent simulations were to be run during the genetic algorithm. Mesh settings were kept consistent across all simulations to avoid introducing artificial variance in the computed electromagnetic quantities. Not having to repeatedly remesh each model also cut computational cost significantly. Because of the axisymmetric assumption, this 2D approach allowed for rapid evaluation of hundreds of resonator configurations, making it well suited to integration with the genetic algorithm discussed in the next section.

As with all finite element simulations, the results are subject to discretisation errors. The use of an extremely coarse mesh reduced computation time but may introduce systematic underestimation of the quality factor and slight distortion of field distributions. However, because all geometries were compared under identical meshing conditions, relative performance trends are expected to remain valid. In addition, uncertainties in material parameters, particularly the dielectric loss tangent and permittivity at GHz frequencies, introduce some variability in Purcell factor estimation.

Finally, each COMSOL model was exported as a **.java** file.

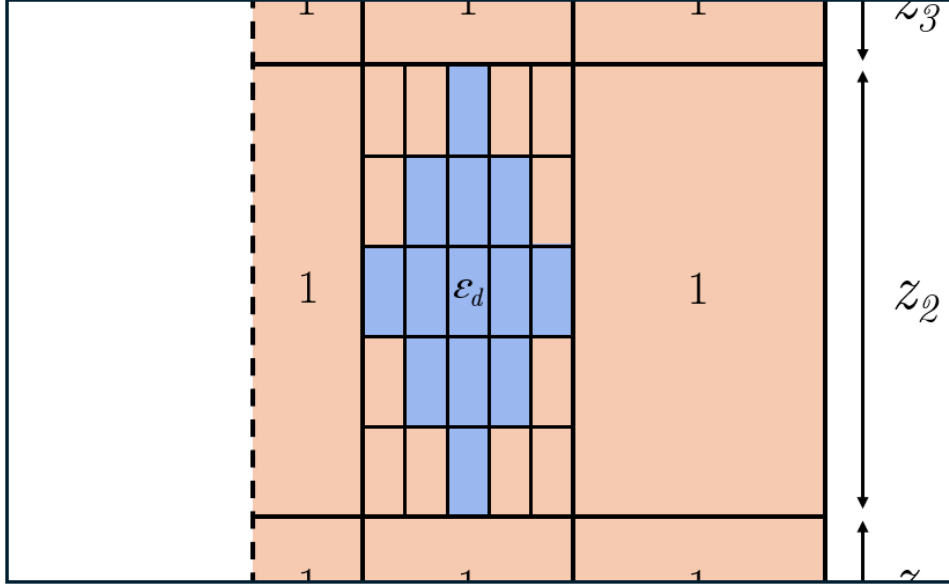


Figure 5.1.2: Illustration of the $N \times N$ subgrid method used to define dielectric geometry within the resonator. Each square element in the grid can be toggled between air ($\epsilon_r = 1$) and dielectric ($\epsilon_r = \epsilon_d$) to form arbitrary shapes. In this example, a 5×5 grid is used to generate a central star-like structure embedded within the air-filled region.

5.2 Java (COMSOL API) – Genetic Algorithm

The primary objective of this project was to find the optimal shape and size of a dielectric resonator that maximises the Purcell factor. To efficiently search the large and discrete design space of possible resonator geometries, a Genetic Algorithm (GA) was implemented and integrated directly with the COMSOL model via the COMSOL Java API. This allowed for automated, iterative variation of the resonator structure based on performance feedback from electromagnetic simulations.

The .java file was then opened using Eclipse IDE (Integrated Development Environment). Eclipse IDE was used for its extensible plug-in system for customising the environment. The COMSOL plug-ins can be referenced within Eclipse IDE in order to run COMSOL simulations without opening the COMSOL GUI (Graphical User Interface).

The GA I wrote in Java (Appendix C) using the Jenetics library[42], can be appended to the .java file and executed. Each candidate solution (i.e. resonator shape) was represented as a 100-bit binary chromosome, corresponding to a 10×10 subgrid over the resonator region. In this representation, each bit determines whether a corresponding domain is filled with the dielectric material (bit = 0) or replaced with air (bit = 1). This effectively enabled the GA to act on a binary material matrix, where the geometry could be adjusted by toggling local permittivity values within a fixed simulation domain.

Before running the full optimisation, the genetic algorithm required an initialised model that was structurally compatible with the 10×10 binary chromosome format. To generate this, a 50/50 configuration was created by applying a checkerboard-like pattern of dielectric and air-filled domains over the resonator region of the baseline model (Figure 5.2.1). This approach ensured that the starting geometry reflected an unbiased, intermediate state between fully filled and fully empty configurations—common practice in GA initialisation to promote exploratory diversity in early generations. Because this modified structure altered the effective dielectric loading of the cavity, its permittivity was then iteratively scaled so that it resonated at the target frequency of 1.45 GHz. This frequency alignment ensured that the GA started from a physically valid configuration and that all subsequent fitness comparisons were made relative to a consistent operating point. Now the GA could be run.

In each generation, the GA evaluated a population of 20 geometries over 250 generations. Standard evolutionary operators were used, including mutation (5%), crossover (80%), and tournament selection also with elite selection. Mutation introduces small, random changes to a geometry’s binary matrix, flipping a few dielectric/air regions, to maintain diversity in the population and avoid premature convergence. Crossover involves combining parts of two parent dielectric geometries (binary matrices) to create new offspring geometries, potentially inheriting favourable features from each parent. Together, these operations allow

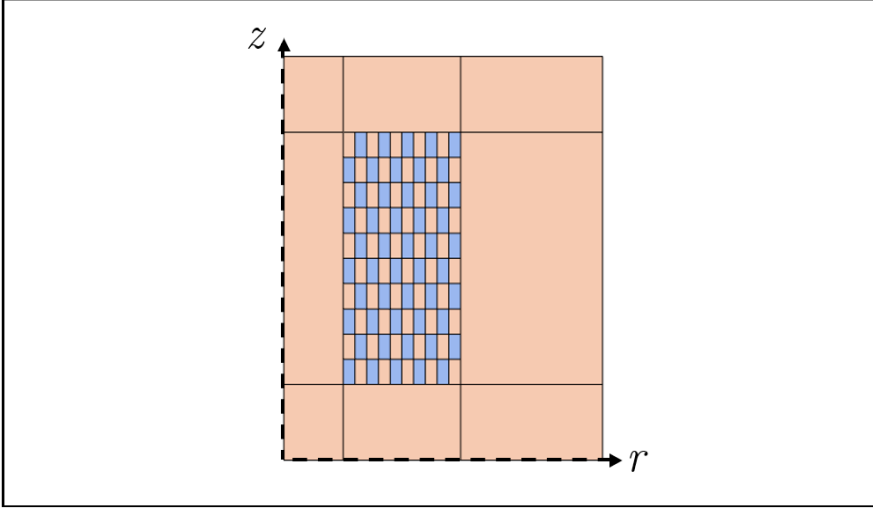


Figure 5.2.1: Initialised configuration of the dielectric subgrid before the genetic algorithm is run. A checkerboard-like 10×10 binary pattern is applied within the resonator region, alternating between air and dielectric elements. This balanced configuration promotes genetic diversity and serves as an unbiased starting point for optimisation.

Table 5.2: The input parameters of the initialised model.

Parameter	Value [mm]
z_1	3.66
z_2	12.3
z_3	3.66
r_1	2.89
r_2	8.59
r_3	15.5

the genetic algorithm to explore a wide design space and progressively evolve higher-performing resonator shapes. Tournament selection describes when two random subsets of a population of 2-5 geometries are created and the individuals with the highest ‘fitness’ will have 80% chance that they will crossover. Elite selection automatically ensures that the geometries with the highest ‘fitness’ carry over into the next generation. An overview of the GA is shown in Figure 5.2.2.

Fitness was assigned based on the calculated Purcell factor (higher the F_P the higher the fitness), which was extracted from each COMSOL run. The simulation results computed the frequency, quality factor, and magnetic flux density, hence V_m can be calculated (see Equation 3.1.2).

To ensure the resonator operated near the target MASER frequency of 1.45 GHz, the GA included an adaptive permittivity scaling loop within each fitness evaluation. This loop aimed to compensate for frequency shifts introduced by changes in geometry by adjusting the global dielectric permittivity, effectively ‘retuning’ each configuration without altering the physical dimensions and therefore remeshing the model.

A frequency shift was first achieved by applying a scalar multiplier α to the real part of the dielectric permittivity ε_r , such that:

$$\varepsilon_{\text{scaled}} = \alpha \varepsilon_r \quad (5.2.1)$$

Since the resonant frequency, f , of a dielectric-loaded cavity is approximately inversely proportional to the square root of the relative permittivity:

$$f \propto \frac{1}{\sqrt{\varepsilon_r}} \quad (5.2.2)$$

The scaling factor required to move from an initial frequency, f_0 , to the target frequency $f_t = 1.45$ GHz is given by:

$$\alpha = \left(\frac{f_0}{f_t} \right)^2 \quad (5.2.3)$$

This allowed the GA to adjust ε_r automatically and bring the resonant frequency of each geometry closer to the desired value, ideally within a ± 10 MHz tolerance. However, in practice, this method encountered issues when implemented. When the scaling factor was applied, the target frequency was rarely achieved. The algorithm would alternate between frequencies too high and too low without the values getting any closer to the target frequency. The cause of this is unknown, this method may be an oversimplification of the relationship between ε_r and f in COMSOL.

To overcome this issue, a Lagrange interpolation method was implemented to estimate the appropriate permittivity scaling that would result in resonance at 1.45 GHz. For three previously simulated data points

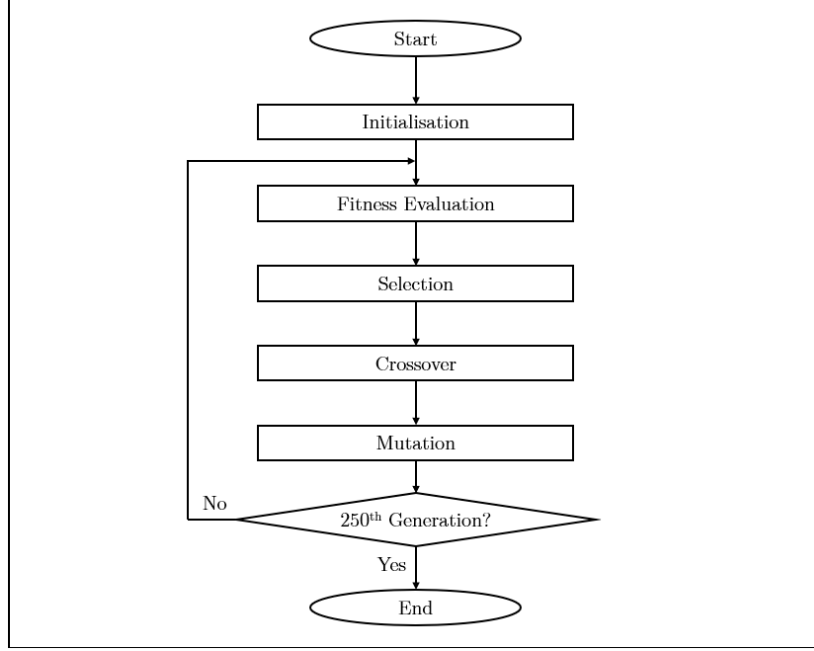


Figure 5.2.2: Flowchart illustrating the genetic algorithm (GA) process used for the optimisation of STO dielectric resonators. The algorithm begins with population initialization, followed by iterative cycles of fitness evaluation, selection, crossover, and mutation. The loop continues until the 250th generation is reached, at which point the algorithm terminates. This structure allows for evolutionary exploration of geometries to improve Q_f/V_m .

(ε_1, f_1) , (ε_2, f_2) , (ε_3, f_3) , the Lagrange polynomial was constructed as:

$$f(\varepsilon) = f_1 \cdot \frac{(\varepsilon - \varepsilon_2)(\varepsilon - \varepsilon_3)}{(\varepsilon_1 - \varepsilon_2)(\varepsilon_1 - \varepsilon_3)} + f_2 \cdot \frac{(\varepsilon - \varepsilon_1)(\varepsilon - \varepsilon_3)}{(\varepsilon_2 - \varepsilon_1)(\varepsilon_2 - \varepsilon_3)} + f_3 \cdot \frac{(\varepsilon - \varepsilon_1)(\varepsilon - \varepsilon_2)}{(\varepsilon_3 - \varepsilon_1)(\varepsilon_3 - \varepsilon_2)} \quad (5.2.4)$$

This polynomial was inverted numerically to estimate the value of ε that would produce $f(\varepsilon) = 1.45$ GHz. This method allowed robust frequency correction without the drawbacks of direct permittivity scaling, enabling consistent evaluation of the fitness function across a diverse set of resonator geometries.

Throughout the evolutionary process, data including the Purcell factor, scaling factor α , and material assignments were logged to .csv files for later analysis. The final output of the algorithm was the dielectric geometry that produced the highest Purcell factor, along with its associated field distribution and frequency characteristics. This approach enabled the discovery of non-obvious geometries with superior field confinement and loss minimisation, effectively bypassing the limitations of manual or purely intuitive design.

Each COMSOL simulation ran in approximately 5 seconds using my personal machine (Intel i7, 16.0 GB RAM). The full GA process, encompassing 250 generations and 20 individuals per generation, took approximately 8 hours of continuous computation.

The final optimised model returned from the genetic algorithm needs to be scaled to return the permittivity values back to the initial values in order for proper comparison with the baseline model.

5.3 Julia – Quantum Cumulants

To evaluate masing performance, the optimised resonator configurations were simulated using **QuantumCumulants.jl**, a Julia package for modelling open quantum systems. The Julia code was adapted with permission from Xipeng Shu, and can be found in Appendix D. The system consisted of a microwave cavity (Fock space) coupled to ten clusters of pentacene molecules, each treated as a five-level spin system using the ClusterSpace abstraction. The system Hamiltonian described coherent spin–photon interactions between the cavity field and molecular transitions (notably between levels 3 and 5), while Lindblad operators captured spontaneous emission, intersystem crossing, spin relaxation, and cavity photon decay. Critically, parameters from the COMSOL simulations, the quality factor Q_f and mode volume V_m , were used to derive the photon decay

rate κ and coupling strength g_{53} in the Julia model. The decay rate was estimated as:

$$\kappa = \frac{\omega_m}{Q_f} \quad (5.3.1)$$

where κ is the cavity decay rate. The coupling strength g_{53} was inferred from V_m , which links the cavity-enhanced emission rate to the local field density (see Equation 3.1.1):

$$g_{53} \propto \sqrt{\frac{1}{V_m}} \quad (5.3.2)$$

This scaling allowed consistent integration of the electromagnetic properties from COMSOL into the quantum dynamics of the spin ensemble. The system was simulated in two phases: optical pumping (0–200 μs) and free evolution (200–500 μs). The time-dependent intracavity photon number $\langle a^\dagger a \rangle$ was extracted and converted into MASER output power via:

$$P_{\text{out}}(t) = \hbar\omega_m\kappa\langle a^\dagger a \rangle(t) \quad (5.3.3)$$

Output power was plotted over time to verify masing behaviour. This model provided a bridge between dielectric design and macroscopic maser performance under realistic room-temperature conditions.

Chapter 6

Results and Discussion

6.1 Model Validation

Both the baseline model, derived from the MASER-in-a-Shoebox dimensions[30], and the initialised model used for genetic optimisation successfully supported resonant modes at approximately 1.45 GHz, operating in the characteristic $TE_{01\delta}$ mode.

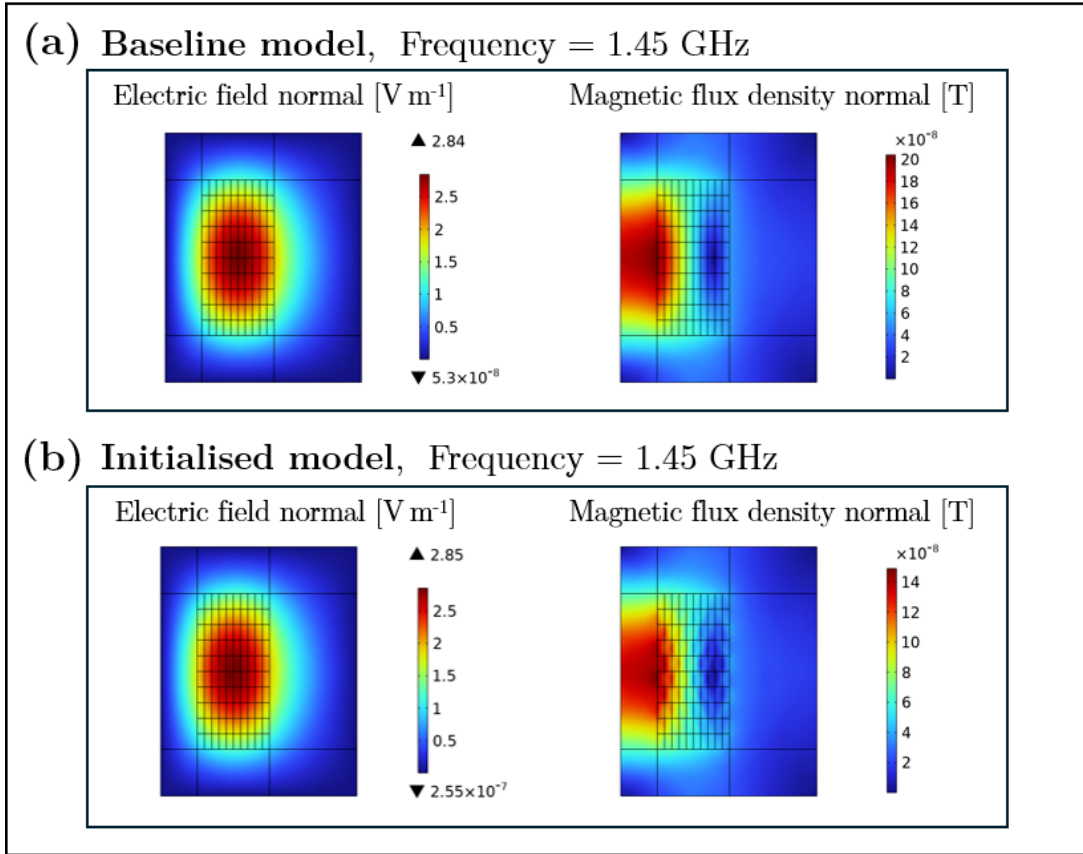


Figure 6.1.1: Comparison of the electric field normal and magnetic flux density normal for (a) the baseline model and (b) the initialised model, both resonating at 1.45 GHz. The baseline configuration replicates dimensions from the MASER-in-a-Shoebox study[30], while the initialised model uses a checkerboard dielectric pattern. Both exhibit strong field confinement within the dielectric region and excitation of the $TE_{01\delta}$ mode.

To validate correct mode excitation and field behaviour, the electromagnetic field distributions of both the baseline and initialised models were examined in COMSOL. In both cases, the $TE_{01\delta}$ mode was successfully identified at a resonant frequency near 1.45 GHz. This mode is characterised by strong azimuthal symmetry in the magnetic field and minimal radial electric field components, consistent with theoretical expectations.

Figures 6.1.1(a) and (b) display the magnetic field intensity distributions for the baseline and initialised models, respectively. In both, the magnetic field is strongly confined within the dielectric resonator region, while the electric field remains weak at the cavity boundaries, confirming that the desired mode has been

correctly isolated. These field patterns provided confidence that the subsequent Purcell factor and quality factor calculations are based on physically valid solutions.

A quantitative comparison between the MASER-in-a-Shoebox-derived baseline model and the initialised model used for the genetic algorithm is presented in Table 6.1. Both models were designed to resonate at 1.45 GHz and employed identical simulation domains, meshing parameters, and boundary conditions. Importantly, these models were computed without including the dielectric loss tangent of the STO. This omission was deliberate, as it allowed for an isolated evaluation of electromagnetic confinement and field distribution without introducing material-dependent damping. However, this also means the reported quality factors represent unloaded or idealised values, not reflective of the actual physical losses in STO.

The mode volume was calculated using the magnetic field distribution and the standard definition described in the Methods section. Subsequently, the $\frac{Q}{V_m}$ was computed for each model. As shown in Table 6.1, the baseline model exhibited a higher Purcell factor than the initialised model, indicating slightly improved field confinement and reduced mode volume.

6.2 Genetic Algorithm Results

Following the establishment of a validated initialised model, the genetic algorithm was executed to optimise the geometry of the dielectric resonator with the objective of maximising the Purcell factor. Over the course of 250 generations, with a population size of 20, the GA iteratively modified the permittivity profile of the resonator using a binary-coded 10×10 subgrid. This allowed the algorithm to explore a wide design space while maintaining the resonant frequency near the target 1.45 GHz. Figure 6.2.1 presents the evolution of the maximum and average $\frac{Q}{V_m}$ over all generations. The rapid increase in both metrics within the first 50 generations indicates efficient convergence toward favourable geometries. Subsequent generations continued to produce incremental gains, with the best-performing geometry emerging around generation 200. This optimised configuration resulted in a substantial increase in Purcell factor compared to both the baseline and the initialised models. Over the whole GA, $\frac{Q}{V_m}$ appears to increase by 600%, which is a very surprising result.

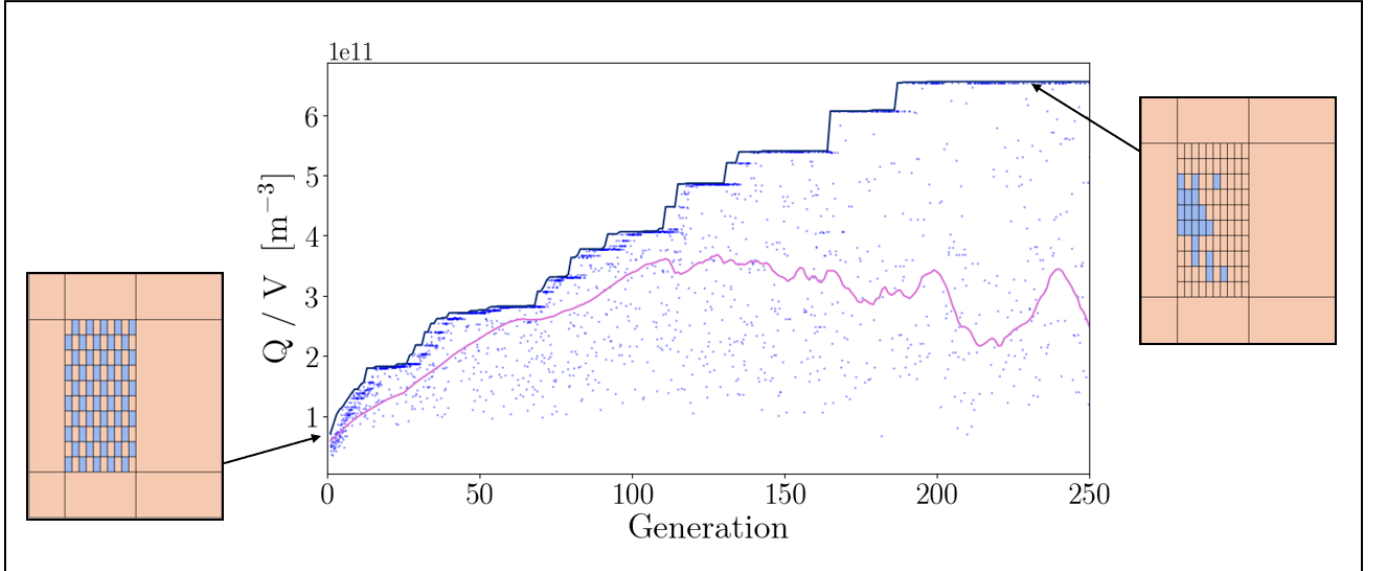


Figure 6.2.1: Evolution of the genetic algorithm over 250 generations. The highest $\frac{Q}{V_m}$ found in each generation is shown by the dark blue line, while the pink line shows the moving average $\frac{Q}{V_m}$. All individual simulation results are plotted as small blue dots. The initial and final dielectric configurations, corresponding to the start and end of the optimisation, are shown alongside the plot.

To better understand how the GA achieved this enhancement, the individual components of the Purcell factor were also tracked. Figure 6.2.2(a) shows the Q_f over all iterations (i.e. every individual evaluated across all generations). The spread of values reflects the diversity of explored geometries. Similarly, Figure 6.2.2(b) displays the V_m over time. A gradual decrease in V_m can be observed across the optimisation process, reflecting a supposed improved electromagnetic confinement within the dielectric structure.

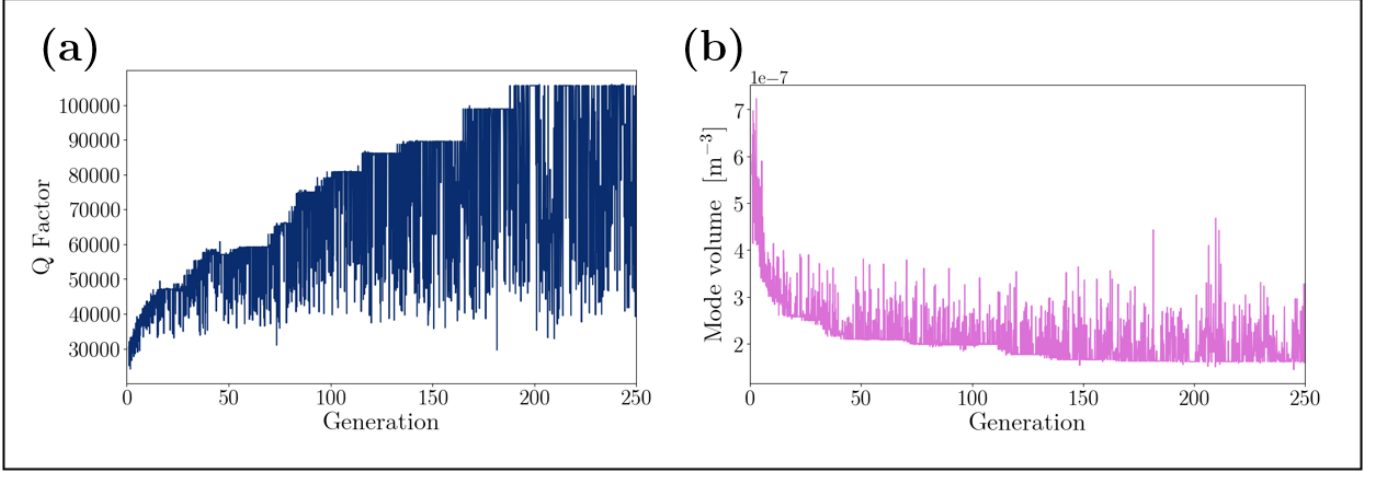


Figure 6.2.2: (a) Quality factor and (b) mode volume as functions of generation number during the genetic algorithm run. As the optimisation progresses, the quality factor generally increases while the mode volume decreases, contributing to the overall improvement in the Purcell factor.

In Figure 6.2.3(a), the resonant frequency of all evaluated geometries is shown. Most newly generated configurations initially resonated at frequencies significantly offset from the 1.45 GHz target. However, the iterative permittivity-scaling loop, incorporating Lagrange interpolation, effectively returned these frequencies to the desired range within a small number of iterations. The convergence of resonant frequencies across the optimisation process demonstrates that the interpolation method was successfully able to correct deviations and ensure consistent evaluation of the Purcell factor across varying geometries. Figure 6.2.3(b) shows the evolution of α (and therefore ϵ) over the course of the GA. α quickly increased to over 1.6 in over 50 generations, corresponding to a global increase in the resonators dimensions. Each simulation started with $\alpha = 1$ before the scaling occurs which is why the graph continually returns to $\alpha = 1$.

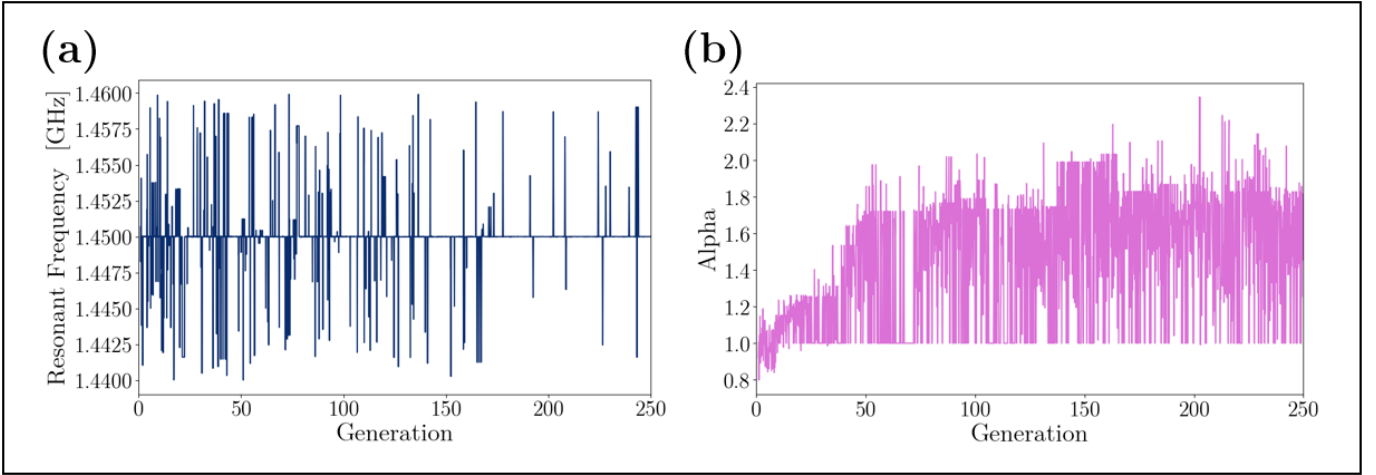


Figure 6.2.3: (a) Resonant frequency and (b) permittivity scaling factor α as functions of generation number during the genetic algorithm run. The frequency plot shows consistent convergence toward the target value of 1.45 GHz, while the increase in α reflects a progressive rise in the scaled relative permittivity values of both air and dielectric regions.

The final optimised configuration, extracted at generation 208, exhibited the highest Purcell factor achieved during the GA run. Its corresponding field distribution, visualised in COMSOL, confirmed strong magnetic field confinement within the dielectric region, consistent with a well-coupled resonant mode. This configuration was then used as the input for the **QuantumCumulants.jl** simulation, enabling the evaluation of practical masing performance.

Optimised model, Frequency = 1.45 GHz

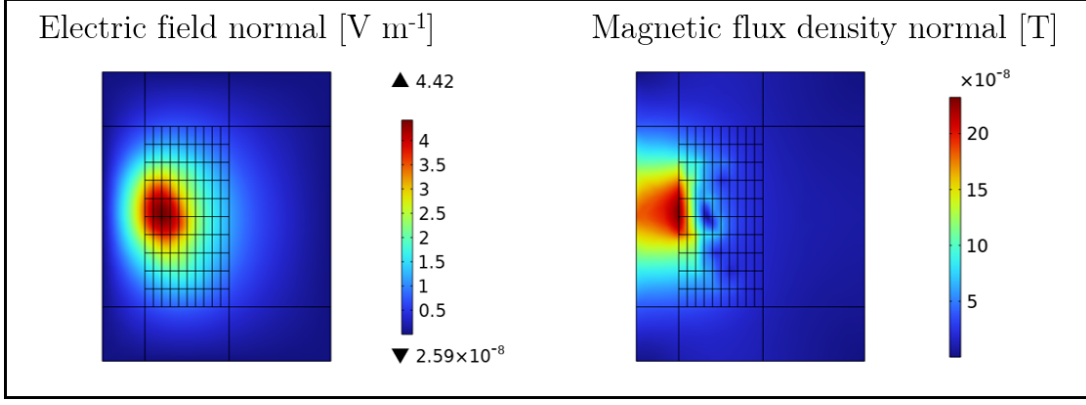


Figure 6.2.4: Electric field normal and magnetic flux density normal distributions of the optimised model at 1.45 GHz. The optimised configuration supports the $TE_{01\delta}$ mode, with strong magnetic field confinement within the dielectric structure.

6.3 Quantum Cumulants Results

To assess how geometric optimisation of the dielectric resonator influences practical masing performance, both the baseline and optimised configurations were simulated using the **QuantumCumulants.jl** package. Each model's calculated quality factor Q_f and mode volume V_m were used to derive the photon decay rate (Equation 5.3.1) and the light-matter coupling strength (Equation 5.3.2), which were then implemented as input parameters in the Julia simulation. These values directly influence the intracavity dynamics and ultimately the masing output power.

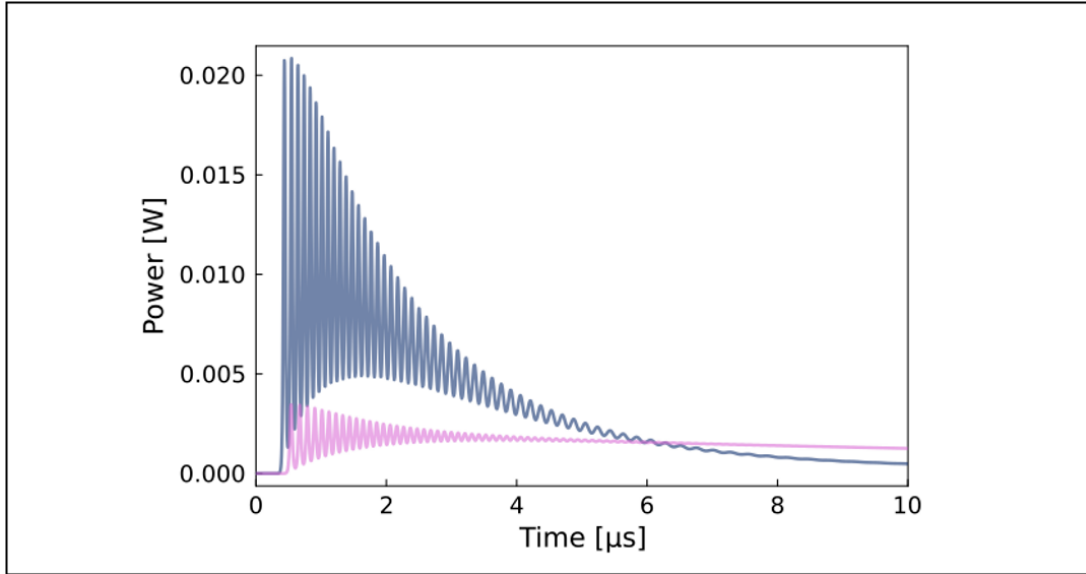


Figure 6.3.1: Simulated output power over time for the optimised configuration (dark blue) and initial configuration (pink). While the optimised resonator shows higher peak power and faster masing onset, these results are purely comparative and do not represent actual output power values. The model serves as a relative performance indicator under idealised conditions.

The simulations tracked the time evolution of the intracavity photon number $\langle a^\dagger a \rangle$, from which the masing output power was calculated via Equation 5.3.3.

As shown in Figure 6.3.1, the optimised configuration produced a noticeably higher peak output power and reached its maximum value more rapidly than the baseline model. This trend is attributed to the enhanced Purcell factor of the optimised geometry, which increases the rate of spontaneous emission into the cavity mode and promotes faster buildup of coherent photons. It is important to note, however, that the output power profiles generated using **QuantumCumulants.jl** are not intended to represent absolute real-world measurements. Instead, they provide a useful visual and comparative framework for evaluating

relative differences in masing performance between configurations. Throughout the simulated free-evolution period, the optimised model consistently outperformed the baseline in terms of sustained output power, indicating more efficient and robust masing behaviour under the idealised conditions used in the model.

These results demonstrate that geometric optimisation, even using a simple pixelated subgrid and binary material assignment, can lead to meaningful performance improvements at the device level. The table below, Table 6.1, summarises the key quantitative metrics extracted from the baseline and optimised models. All values reflect the unloaded (lossless) simulations as the dielectric loss tangent was excluded from both COMSOL and Julia simulations to isolate geometric contributions.

Table 6.1: A comparison of the three losses models.

	Frequency [GHz]	Q_f	V_m numerator	V_m denominator [m ³]	Q_f / V_m [m ⁻³]
Baseline	1.45	21070	5.539×10^{-9}	0.02638	1.003×10^{11}
Initialised	1.451	29580	7.845×10^{-9}	0.01403	5.290×10^{10}
Optimised	1.449	1.382×10^5	1.247×10^{-8}	0.03436	3.806×10^{11}

6.4 Limitations of Results

While the results presented demonstrate clear improvements in resonator performance through geometric optimisation, several important limitations must be acknowledged, most notably, the simplified treatment of material losses. In all COMSOL simulations used throughout this project, the dielectric loss tangent of STO was deliberately omitted to try and retain simplicity and increase the speed at which the GA runs. Ultimately, this resulted in idealised and unrealistic performances. In practical implementations, STO exhibits a non-negligible loss tangent (on the order of 10^{-4} to 10^{-3}), which would substantially reduce the real-world Q_f , especially at GHz frequencies. The equation for the total Q factor considering all major methods of loss can be seen in Equation 6.4.1.

$$\frac{1}{Q_{\text{tot}}} = \frac{1}{Q_{\text{ohm}}} + \frac{1}{Q_{\text{diel}}} + \frac{1}{Q_{\text{rad}}} \quad (6.4.1)$$

Where Q_{ohm} is the Q_f of the cavity walls, Q_{diel} is the Q_f of the dielectric (determined by the loss tangent, $\tan \delta$), and Q_{rad} is the Q_f of radiative losses. This method assumes $1/Q_{\text{tot}} = 1/Q_{\text{ohm}}$.

This omission affects the physical accuracy of both the COMSOL and Julia simulations. In COMSOL, losses are only accounted for via the impedance boundary condition applied to the copper cavity walls, which introduces conductive losses through surface current dissipation. Additionally, any radiative losses due to imperfect field confinement are inherently captured by the finite simulation domain and open boundary conditions. However, dielectric losses, often the dominant loss channel in high-permittivity materials like STO, are not represented, and therefore the reported Purcell factors are overestimated relative to real-world performance.

In fact, when the optimised configuration was simulated with a dielectric which included a loss tangent. The gains in Purcell factor were found in lossy conditions and the Purcell factor was lower than that of the baseline model. Therefore, this model would not outperform the baseline model in experimental conditions.

6.5 Loss-included Models

To move toward a more physically realistic optimisation, an updated version of the genetic algorithm (GA) was developed with the aim of including the loss tangent of the dielectric material. This was implemented by assigning a complex relative permittivity (in the $\tan \delta$ column of Appendix A.1) to the dielectric regions within the COMSOL model, introducing an imaginary component proportional to the specified loss tangent.

In principle, this would allow the GA to optimise not just for electromagnetic confinement but also for geometries that minimised overall dissipative loss. However, this modification introduced a significant computational challenge: in the presence of realistic dielectric loss, the solver consistently found incorrect modes. Many configurations either failed to converge to a valid mode or converged to unintended resonant

solutions with poor confinement and irrelevant field distributions. This instability made the fitness function unreliable, preventing the GA from effectively progressing. As a result, the optimisation loop was unable to identify meaningful high-performing geometries under lossy conditions. This is all highly likely caused by an unidentified bug in my code.

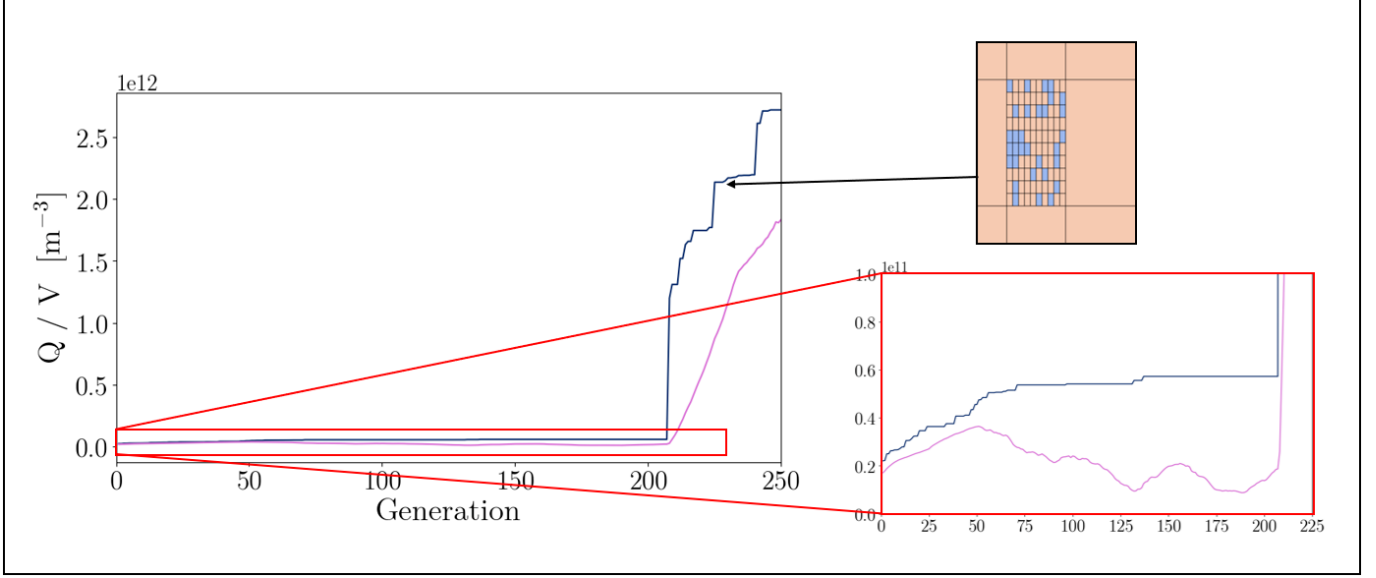


Figure 6.5.1: Progression of the genetic algorithm under lossy conditions. The dark blue line shows the highest $\frac{Q}{V}$ recorded at each generation, while the pink line represents the average $\frac{Q}{V}$ across all simulations. The inset zooms in on the slow initial progress of the algorithm, followed by a sudden breakthrough in performance after approximately 200 generations. The final configuration is also shown.

The configurations returned from the failed model were extracted to try to identify where the root of the issue was. In Figure 6.5.1, even though the F_P does increase significantly towards the end of the GA, this is due to mode switching. This is confirmed by Figure 6.5.2, which shows the resonator is not longer operating in the $TE_{01\delta}$ mode. Mode switching during the GA is a problem as the model loses its consistency, so any results are invalid.

Failed model, Frequency = 1.45 GHz

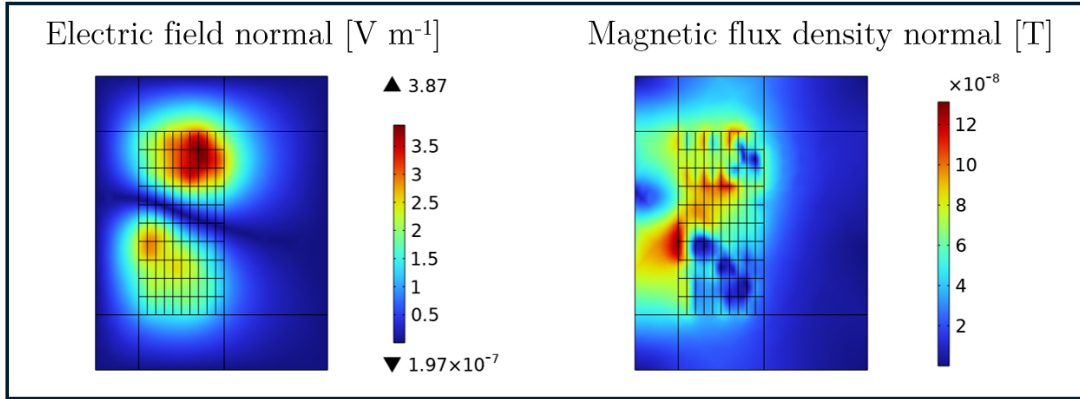


Figure 6.5.2: Electric field normal and magnetic flux density normal for a failed configuration that resonated at 1.45 GHz. Its shape can be seen in 6.5.1. The magnetic flux density shows poor confinement within the dielectric region, with significant energy leakage into surrounding areas. This indicates weak field–material interaction, resulting in a low Purcell factor and ineffective masing behaviour.

To verify that these failures were not due to the unfeasibility of enhancing the Purcell factor in lossy configurations, manual simulations were performed in order to find more optimal geometries. Several configurations were found to exhibit Purcell factors higher than that of the baseline MASER-in-a-Shoobox model. One result is displayed in Figure 6.5.3.

These results confirmed that performance improvements were physically achievable even with realistic losses. A comparison to the baseline model is shown in Table 6.2.

Manually improved model, Frequency = 1.45 GHz

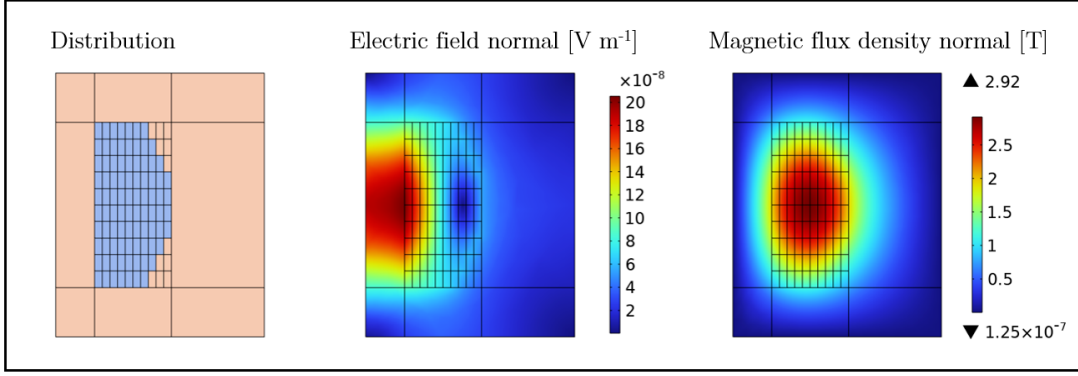


Figure 6.5.3: Manually optimised configuration demonstrating improved field confinement. The distribution shows the dielectric domains (blue).

Table 6.2: Comparison of baseline and manually optimised configurations in the presence of dielectric loss.

	Frequency [GHz]	Q_f	V_m numerator	V_m denominator [m ³]	Q_f / V_m [m ⁻³]
Baseline	1.45	6548	5.539×10^{-9}	0.02638	3.119×10^{11}
Manual	1.451	6988	5.618×10^{-9}	0.02672	3.324×10^{11}

Due to time constraints of the project, a full redesign of the frequency correction procedure could not be implemented. However, a feasible solution was proposed: frequency tuning with Q-neutral permittivity adjustment.

This method still involves adjusting the global permittivity to bring the resonant frequency to 1.45 GHz, but critically it does so without altering the total quality factor Q_{total} of the dielectric material. This would require incorporating the filling factor, η , which quantifies the fraction of the electromagnetic field energy stored within the dielectric:

$$\eta = \frac{\int_{\text{dielectric}} \epsilon |\mathbf{E}|^2 dV}{\int_{\text{total}} \epsilon |\mathbf{E}|^2 dV} \quad (6.5.1)$$

Given this, the total quality factor Q of the system can be expressed as:

$$\frac{1}{Q_{\text{tot}}} = \frac{\eta}{Q_{\text{diel}}} + \frac{1}{Q_{\text{other}}} = \eta \tan \delta_{\text{diel}} + \frac{1}{Q_{\text{other}}} \quad (6.5.2)$$

Where Q_{other} represents other losses such as conduction (Q_{ohm}) and radiation (Q_{rad}).

Q_{tot} needs to remain fixed and while the frequency is scaled to the target frequency.

By decoupling frequency correction from loss scaling, the proposed method allows more stable exploration of the design space, preserving the integrity of the fitness function and enabling meaningful optimisation even under lossy conditions. Implementing this strategy would require integrating filling factor calculations into the GA loop and defining the dielectric loss independently of the real permittivity scaling. While more complex, it offers a promising path forward for loss-aware resonator optimisation.

6.6 Additional Further Work

Several other opportunities for further development remain. One such improvement is the implementation of a ‘surface energy’ factor within the genetic algorithm’s fitness evaluation. The current framework allows for isolated dielectric domains with disconnected regions that may not form a coherent 2D structure. While such configurations can yield high Purcell factors, they will be physically difficult to manufacture. Introducing a ‘glueing’ penalty or reward term could encourage the evolution of continuous, manufacturable geometries by favouring designs that form a single contiguous resonator body. Whether such isolated domains could

be made physically viable remains an open question. One speculative approach might involve embedding discrete dielectric regions within a surrounding resin that has electromagnetic properties closely matching those of air. This would theoretically preserve the field distribution observed in simulation, but such materials would require extremely low loss tangents and negligible permittivity. In practice, enforcing coherent shapes through algorithmic constraints or penalty terms is likely a much more feasible path forward.

Another avenue for future work is expanding the optimisation into full 3D modelling. While this study relied on COMSOL's 2D axisymmetric mode, due to its efficiency and alignment with cylindrical geometries, more complex or asymmetric dielectric shapes may yield even greater improvements in Purcell factor. Accessing these designs requires abandoning rotational symmetry and operating within a full 3D simulation framework. The increased computational load associated with 3D simulation would necessitate a more efficient integration pipeline than was available in this project.

This may be implementable using the COMSOL LiveLink™ for MATLAB® that represents an improvement over the Java API used in this study. MATLAB®'s powerful numerical libraries, intuitive syntax, and data handling capabilities are far better suited to genetic algorithms and other optimisation routines. MATLAB®'s environment would allow easier integration of adaptive mutation strategies, multi-objective fitness functions, and visualisation of convergence trends. Unfortunately, the MATLAB® LiveLink™ package requires an additional paid licence, which was not purchased for this project due to its high price and initial unawareness of its benefits. As a result, the project proceeded using the COMSOL Java API, which, while functional, proved more cumbersome and less flexible for advanced algorithmic control.

Looking further ahead, machine learning (ML) techniques could offer an extension to this optimisation framework. A trained surrogate model, for instance, could rapidly estimate Purcell factors for proposed geometries without requiring a full COMSOL simulation at each step. This could accelerate convergence and enable exploration of even larger and more complex design spaces. Combining machine learning with LiveLink™ for MATLAB® would allow for powerful hybrid approaches, such as reinforcement learning-based geometry generation or Bayesian optimisation over a learned performance surface.

6.7 Scientific and Practical Implications

From a scientific perspective, this outcome reinforces a common challenge in cavity QED and microwave engineering: the trade-off between field confinement and loss minimisation. High confinement geometries often exacerbate losses unless materials are carefully chosen or cooling is applied. This is consistent with trends in the literature, where many high-Q devices rely on cryogenic operation, ultra-low-loss dielectrics like sapphire, or careful modal engineering (e.g., whispering gallery or anapole modes) to suppress dissipation[43]. In terms of data quality, uncertainties stem not from numerical error, since COMSOL's eigenfrequency solver is highly accurate. But rather from physical assumptions such as perfect material behaviour, and the neglect of thermal effects.

This project has highlighted the potential of automated shape optimisation for electromagnetic structures using genetic algorithms, offering a computational route to discovering non-intuitive geometries. From an innovation and industry perspective, this method, if made loss-aware, could support the development of miniaturised, room-temperature MASERs with practical applications in quantum sensing, deep-space communication, and ultra-low-noise amplifiers. However, reaching this point will require further integration of material science (for low-loss dielectrics) and experimental feedback (to validate optimised geometries).

In summary, while the project successfully demonstrates an increase in Purcell factor under idealised conditions, the realistic impact of loss limits the viability of the discovered designs without further refinement. These findings are therefore preliminary but insightful, pointing the way to more physically accurate optimisation strategies in the next generation of MASER design.

Chapter 7

Conclusions

This project set out to investigate whether the shape of a dielectric resonator can be optimised to enhance the Purcell factor, and consequently improve the performance of a solid-state MASER operating at room temperature and zero magnetic field. Using COMSOL Multiphysics®, COMSOL API and Java, a genetic algorithm (GA) was implemented to iteratively optimise the geometry of a SrTiO₃-based resonator. The primary objective was to maximise the ratio of quality factor (Q_f) to mode volume (V_m), which are the main metrics that determine the Purcell factor of the resonator and is very important in determining the efficiency of microwave amplification.

The first significant outcome of this work was the successful creation and validation of an initialised geometry that exhibited strong TE_{01δ} mode confinement at the target frequency of 1.45 GHz. This was achieved by modifying the original MASER-in-a-Shoebox dimensions and tuning the resonator using geometric adjustments, mimicking the real-world tuning process via adjustable cavity ceilings. This model provided a baseline for optimisation and ensured that all GA-evaluated geometries could be fairly compared.

The GA produced a series of candidate geometries, with the best-performing structure demonstrating a marked improvement in Purcell factor compared to the baseline model. While this improvement was obtained under idealised (lossless) conditions, it confirmed that purely geometrical changes can enhance the electromagnetic confinement properties of the resonator. Julia-based simulations using **QuantumCumulants.jl** showed that this translated into increased output power under strong coupling assumptions, although these results are illustrative rather than predictive, serving as a visual tool to compare performance between the baseline and optimised models. However, when the dielectric loss of SrTiO₃ was introduced into the simulation, the GA was very slow at discovering more optimal configurations and exhibited mode switching. Thus making any results invalid. After existing that configurations with a higher Purcell factor existed, the results were further examined to determine where the root of the error was. It was ultimately concluded that the code contained a bug.

This project has shown that resonator geometry can play a meaningful role in enhancing MASER performance. Despite the results being preliminary, they (and the code produced) should hopefully serve as a framework for future investigations into this area of research. The Java-based GA proved a suitable prototype however more sophisticated numerical control is required as Java lacks the native libraries and numerical tools. Hence, the migration to COMSOL's LiveLink™ for MATLAB® environment has been proposed. This environment can utilise the more flexible and numerically robust to implement new optimisation techniques, such as machine learning or simulations in 3D.

In summary, this project has demonstrated that resonator geometry significantly affects key MASER performance metrics. While lossless simulations provide a useful sandbox for testing ideas, realistic modelling and more advanced optimisation tools will be essential for translating these insights into deployable MASER technologies.

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